

Dynamic Feature Identification in Echocardiograms

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Abstract—Mitral valve regurgitation is the most common cardiovalvular disease worldwide and relies heavily on echocardiographic imaging for diagnosis and treatment planning. However, accurate interpretation of echocardiogram videos is challenging and requires extensive clinical expertise. This project proposes a digital signal processing framework for automatic identification of mitral valve structures in echocardiogram videos. Using open-source echocardiographic datasets, the system aims to detect, mask, and label dynamic cardiovascular features such as the mitral valve leaflets. The proposed approach leverages spatial edge detection, temporal correlation, and motion-based masking techniques implemented in MATLAB and Python. This work seeks to demonstrate how classical signal processing methods can assist cardiac sonographers by improving the speed and consistency of echocardiographic analysis.

Index Terms—Echocardiogram, Mitral Valve, Computer Vision, Digital Signal Processing

I. INTRODUCTION

Mitral valve regurgitation (MR) affects more than 175 million people globally and can cause symptoms such as fatigue, shortness of breath, and progression to heart failure. Unfortunately, typical ways to cure MR require open-heart surgeries, which are high-risk and invasive. Special surgeries that are minimally invasive include Transcatheter Edge-to-Edge Repair (TEER). However, these surgeries require a more thorough inspection, accurate diagnosis, and treatment planning before the performance. These diagnosis processes rely on echocardiography, including both two-dimensional and Doppler imaging, further depend on real-time echocardiographic guidance during procedures.

Despite its clinical importance, echocardiogram interpretation remains difficult and time-consuming, requiring years of experience from trained cardiac sonographers. Variability in image quality, patient anatomy, and cardiac motion further complicates analysis. This project aims to reduce these challenges by developing a digital signal processing (DSP) pipeline capable of automatically identifying dynamic mitral valve structures in echocardiogram videos, thereby assisting clinicians during diagnosis and treatment, and increasing the market penetration rates of minimally invasive TEER surgeries.

To design the DSP pipeline, this project successfully identifies features such as the left ventricle (LV) borders and the Mitral Valve (MV) leaflets while they are in motion. This project uses Stanford's AIMI Shared Dataset as the objects

for DSP. The echocardiogram videos are de-identified and are taken in the frame for an Apical 4-chamber view [3]. The anatomical structure is demonstrated in Fig. 4.

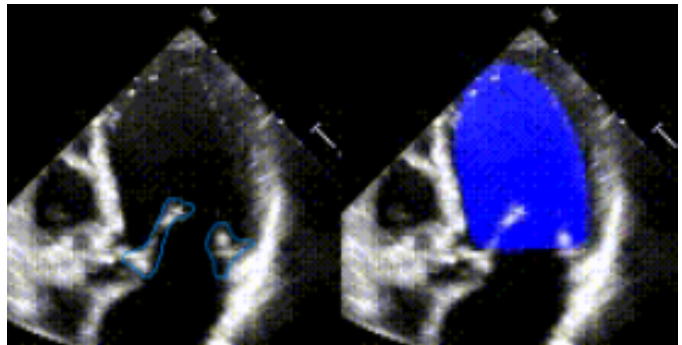


Fig. 1: Left: The mitral valve leaflets, demonstrated and circled by faintly blue markings. Right: The left ventricle, demonstrated with a blue mask [3].

II. TECHNICAL APPROACH: LEFT VENTRICLE BORDER IDENTIFICATION

This section presents a pipeline for identifying the left ventricle in echocardiograms frame-by-frame. The proposed approach relies on classical image filtering, morphological segmentation, and geometric priors, without the use of learning-based models or annotated training data. The pipeline consists of four main stages: image denoising and contrast enhancement, component segmentation via watershed analysis, heuristic component merging, and ventricle identification through a weighted centroid, area, and shape error metric.

A. Cleaning the Image

We first use multiple filters to pool and reduce the feature map size, for the individual features to be more invariant and easily identifiable. A median filter was used to remove speckled bright pixels in the dark regions of the heart image [5]. The image contrast was then doubled. While the resulting image appeared crisp, many of the walls separating dark regions contained gaps, causing separate dark regions to appear as a single connected component.

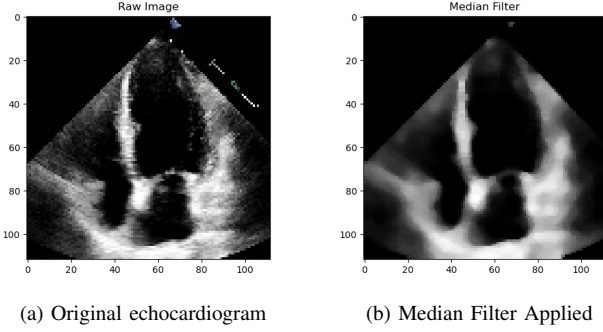


Fig. 2: Median filter applications on echocardiogram frames.

In addition, the left ventricle contained a valve that separated it into an upper and lower half. These two image properties complicated identification of the left ventricle. To address this, a simple vertical 3×7 Gaussian filter was applied to fill small wall gaps and remove ventricle valves [6].

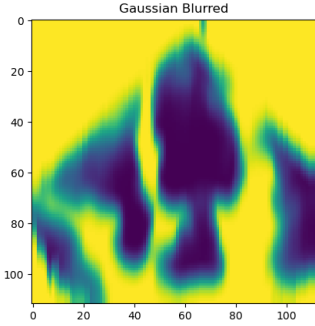


Fig. 3: Applying a gaussian filter for an extra pooling layer.

Pixel values ranged from 0 to 255. The image was thresholded to create a binary mask, where pixel values below 90 were set to 0 and values exceeding 90 were set to 255. The resulting image resembled a silhouette of the components in the heart image.

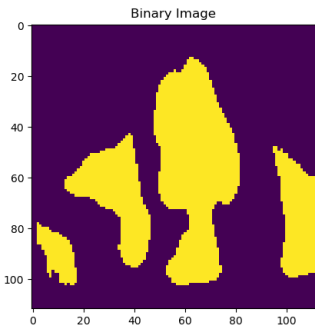


Fig. 4: Binary filter for the final outline of large regions in the frame. The 3rd feature from the left is the left ventricle.

B. Identifying Components

Despite thresholding, components often overlapped with each other. To separate overlapping components, the watershed algorithm was used. As preprocessing for watershed, a distance transform was applied to the binary image, where the value of each bright pixel was defined as its distance to the nearest dark pixel.

Local maxima in the distance map were identified using a simple thresholding approach: pixels with values greater than the average half-width of the left ventricle were marked as maxima. The half-width was chosen to ensure that points near the center of the left ventricle would be reliably detected as maxima. These maxima served as the initial labels for the watershed algorithm.

An intensity map for the watershed algorithm was then created by inverting the distance map and applying a square root to each pixel value. This produced an intensity map with lower values at the centers of components and higher values at the edges.

The watershed algorithm was run using the generated labels and intensity map. Neighbors of the labeled points were identified and placed into a stack. At each step, the point with the lowest intensity-map value was removed from the stack, and the labels of its neighboring pixels were examined. If only one label was present, the point was assigned that label and its neighbors were added to the stack. If two or more labels were present, the point was determined to lie between components and was assigned to a border label. This process continued until the stack was empty. Visually, this procedure resembled water filling basins defined by the intensity map. The resulting image successfully labeled individual components despite overlap [7].

C. Merging Components

Often, the valve in the left ventricle separated the upper portion of the ventricle from the lower portion. To address this, new components were created by combining any component lying directly below the centroid of another component. These combined components did not replace the original components — they were just added to the list of components.

D. Identifying the Left Ventricle

The left ventricle needed to be identified among the multiple components produced by the watershed algorithm. The left ventricle exhibited three consistent properties across the heart images: its centroid was approximately at the center of the image, it had an average pixel area of approximately 2500 pixels, and it exhibited a paddle-like shape. Errors between these expected properties and each component's properties were computed, weighted, and summed. The component with the lowest total error was identified as the left ventricle.

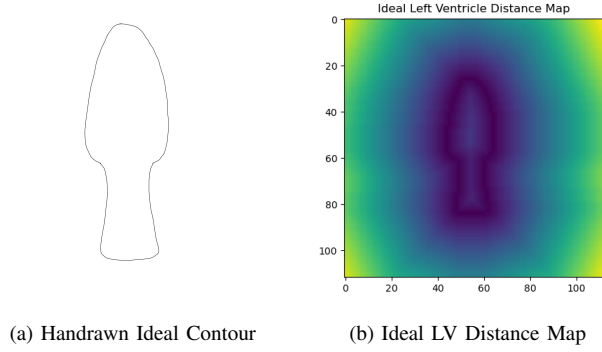


Fig. 5: Hand-drawn Ideal Contours and Maps

An ideal left ventricle shape was manually drawn. Each component was then translated, rotated, and scaled to match the ideal left ventricle's centroid, orientation, and area. Contours were extracted for both the ideal ventricle and the component. A distance transform was applied to the ideal ventricle contour image, followed by computation of the gradient at each point in the distance map [8] [9]. Using this gradient map, each point on the component contour was iteratively moved toward lower values in the distance map, corresponding to the ideal ventricle contour. Iterations continued until the average distance between component contour points and the nearest ideal ventricle contour points reached a threshold. The number of iterations required for convergence was defined as the shape error.

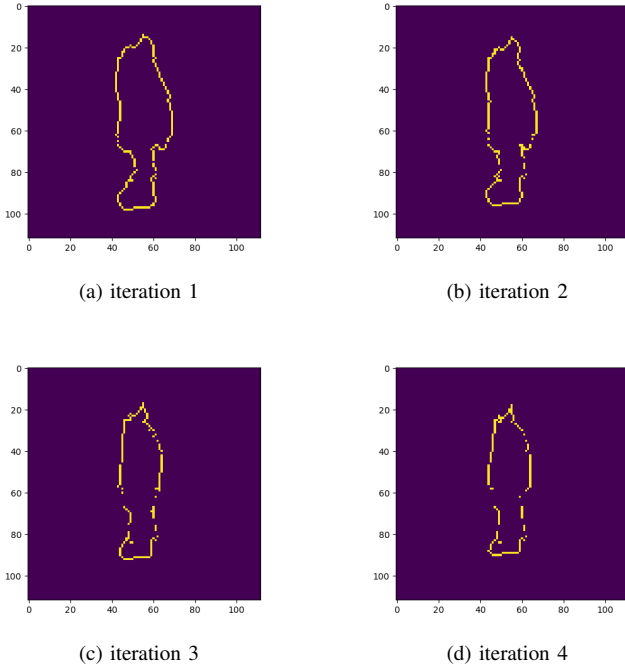


Fig. 6: Iterations of the contour maps.

The centroid location error was computed as the Euclidean

distance between the component centroid and the average left ventricle centroid, demonstrating Fig. 6. The pixel size error was computed as the absolute difference between the component's pixel area and the average left ventricle area. The shape error was more complex to compute.

After converging the final iterations, we obtain a binary mask of the LV region, demonstrated in Fig. 7.

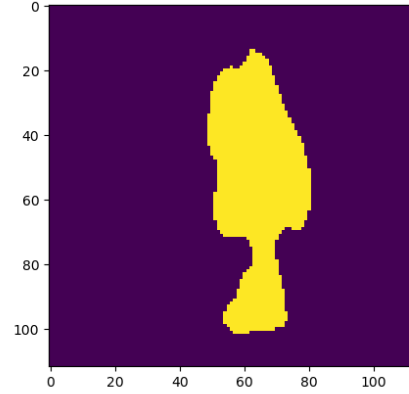


Fig. 7: Final Identified Left Ventricle Outline

III. TECHNICAL APPROACH: MITRAL VALVE LEAFLET IDENTIFICATION

This section presents general technical approach to a fully signal-processing-based framework for identifying mitral valve leaflet motion in echocardiogram videos. The proposed approach relies exclusively on temporal motion cues, spatial priors, and morphological structure, without the use of learning-based models or annotated training data. The pipeline consists of four main stages: preprocessing, motion energy estimation, motion-supported leaflet localization, and frame-wise leaflet pixel extraction.

A. Preprocessing

The preprocessing steps standardize image statistics across frames and videos, ensuring that subsequent motion estimation is driven by relative temporal changes rather than global brightness variations or acquisition-dependent intensity offsets. Each input echocardiogram video is processed frame by frame. Frames are first converted to grayscale (if necessary) and cast to single precision. To suppress high-frequency speckle noise inherent to ultrasound imaging, a small Gaussian filter is applied to each frame. Following denoising, frames are normalized to zero mean and unit variance and subsequently rescaled to the range $[0, 1]$.

B. Motion Energy Estimation

Regions of dynamic activity are identified by computing a global motion energy map across the video sequence. Motion is estimated using temporal differencing between consecutive frames, with an additional backward-difference suppression term to reduce oscillatory artifacts. Negative responses are

clipped to zero, and motion contributions are accumulated across all frames to emphasize persistent motion patterns.

Formally, the motion energy at pixel location (x, y) is computed as:

$$M(x, y) = \sum_t \max \left(|I_{t+1}(x, y) - I_t(x, y)| - \lambda |I_t(x, y) - I_{t-1}(x, y)|, 0 \right) \quad (1)$$

where I_t denotes the preprocessed frame at time t and λ is a fixed attenuation factor. The resulting motion energy map is then spatially smoothed using a Gaussian filter and normalized to $[0, 1]$.

This accumulated motion representation highlights regions exhibiting consistent mechanical activity over the cardiac cycle while suppressing transient noise and isolated fluctuations.

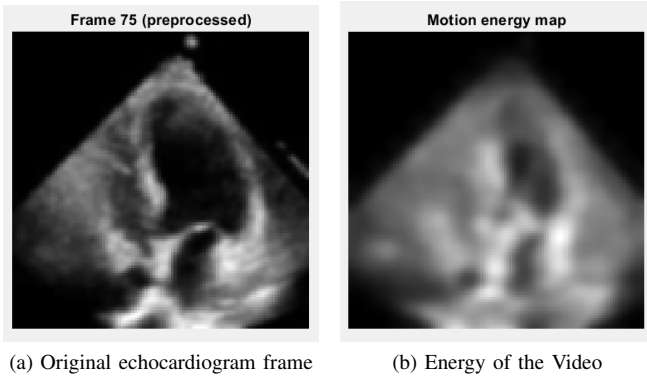


Fig. 8: Median filter applications on echocardiogram frames.

C. Motion-Supported Leaflet Localization

Mitral valve leaflet motion is localized by combining the global motion energy map with a fixed geometric prior and local spatial aggregation. A rectangular band corresponding to the basal region of the imaging plane is defined to restrict analysis to anatomically plausible locations. This geometric constraint reduces false positives arising from unrelated moving structures elsewhere in the image.

Within this band, local motion support is computed by convolving the motion energy map with a uniform spatial kernel. This aggregation step reinforces spatially coherent motion patterns and enables weak but correlated motion signals—to be retained. Thresholding is applied to the motion support values based on their mean and standard deviation within the geometric band, rather than on raw motion intensity.

The resulting binary mask defines a coarse, time-invariant leaflet motion region that captures the spatial extent of mitral valve dynamics. This mask is replicated across all frames and serves as a gating region for subsequent fine-scale leaflet extraction.

D. Leaflet Pixel Extraction

Fine-grained leaflet structures are extracted on a frame-by-frame basis within the motion-localized region. For each

frame, two complementary sources of evidence are computed: normalized intensity values and local motion consistency. Intensity provides contrast-based information, while motion consistency is estimated via spatial averaging of the motion region mask to emphasize locally supported structures.

These two cues are fused using a weighted linear combination to form a continuous leaflet likelihood map. Adaptive thresholding based on the distribution of fused scores within the motion region is then applied to obtain a binary leaflet mask. Small isolated components are removed, and morphological skeletonization is used to enforce thin, line-like structures consistent with valve anatomy. A controlled dilation step is finally applied to improve visual continuity without introducing excessive blurring or over-segmentation.

This stage converts coarse motion localization into anatomically plausible leaflet traces that evolve smoothly across frames.

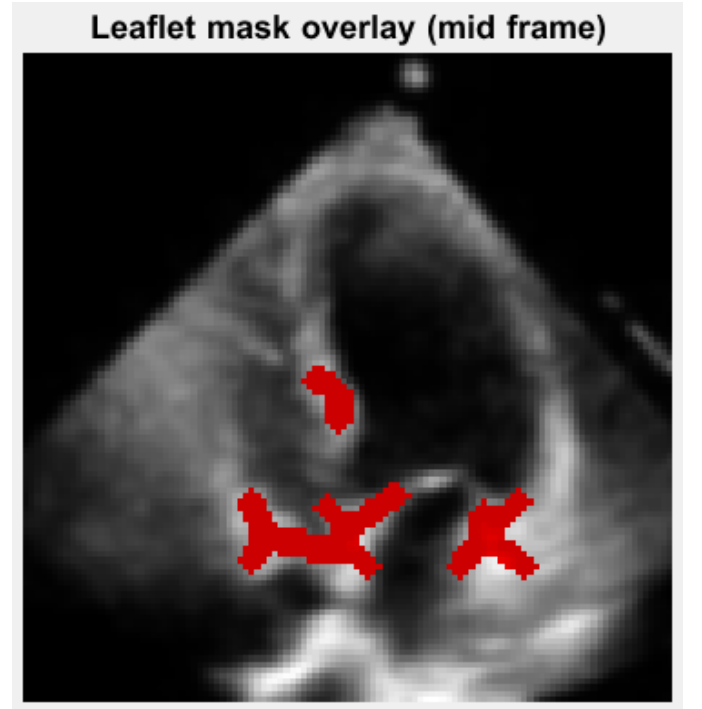


Fig. 9: Resultant Mask of the leaflets.

IV. EXPERIMENT

This section evaluates the proposed algorithms using echocardiography data from the Stanford AIMI Dataset [3]. The left ventricle (LV) identification algorithm is assessed on individual image frames, while the mitral valve (MV) identification algorithm is evaluated on temporal video sequences. Experimental results and qualitative analyses are presented to demonstrate the performance of each method.

A. Left Ventricle Identification

The algorithm successfully identified the left ventricle when the walls separating heart components were well defined, the

heart valve was absent, and the contrast between dark and bright regions was high. Fifty images of the heart were tested on.

For each image, contours were manually drawn around the left ventricle. Separately, the left ventricle identification algorithm was run on the 50 images to produce calculated contours. The drawn contours and the calculated were both filled to create left ventricle images. In these images, the left ventricle areas had pixel values of 1, while the background had pixel values of 0. For each heart image, the drawn left ventricle image was subtracted by the calculated left ventricle image. The non-negative values in the resulting image were summed and divided by the image pixel size to yield the percentage of pixels that were misclassified in the image. The average percentage of misclassified pixels for the 50 heart images turned out to be 15.0%, which shows an 85% accuracy.

However, the algorithm often failed to capture the tip of the left ventricle because this region had low contrast relative to the surrounding walls. During thresholding, pixels at the ventricle tip frequently fell below the threshold value of 90 and were classified as background, resulting in the omission of the tip from the binary mask.

The algorithm also struggled to capture the portion of the left ventricle located below the valve, as the valve often split the ventricle into two separate components. This issue was partially addressed by applying a vertical Gaussian filter to blur the valve and by merging components located above and below one another. Nevertheless, these corrections were not always sufficient. In particular, when the left ventricle was oriented away from the vertical axis, the valve was not perfectly horizontal, reducing the effectiveness of the vertical Gaussian filter. Additionally, the lower portion of the ventricle did not always align directly beneath the centroid of the upper portion, preventing the two components from being merged.

In some cases, the opposite problem occurred, where the algorithm incorrectly merged the left ventricle with a neighboring component. This typically happened when the walls between components contained large gaps or were faded and thin. Under these conditions, the binary image merged components across the weak wall. When the distance transform was applied, a single maximum was identified between the merged components. Because this maximum served as the label for the watershed algorithm, the merged components were assigned the same label, making it impossible to separate them in subsequent steps. Example experiments are presented in the following few figures.

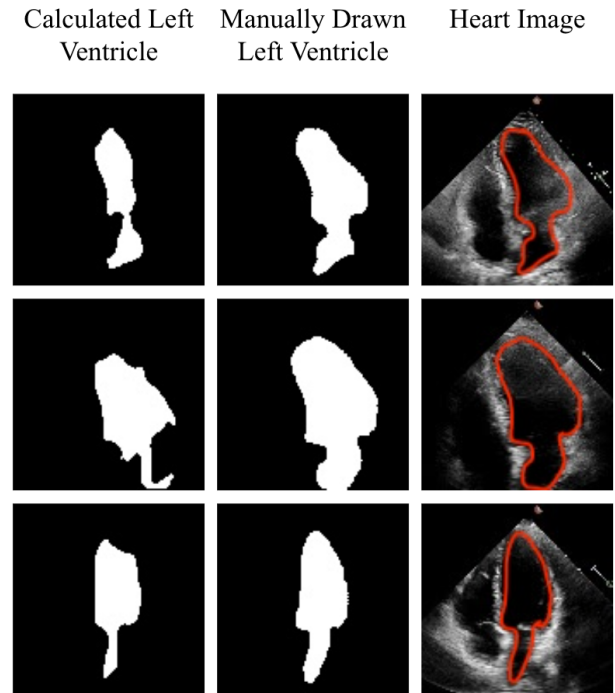


Fig. 10: Examples of Well Calculated Left Ventricle (left) and Drawn Left Ventricle (right) Pairs

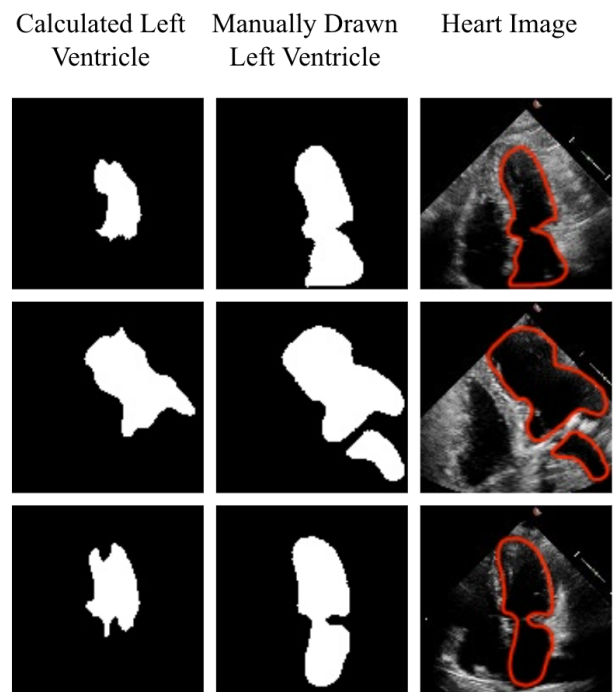


Fig. 11: Examples of Missing Lower Portions in the Calculated Left Ventricle

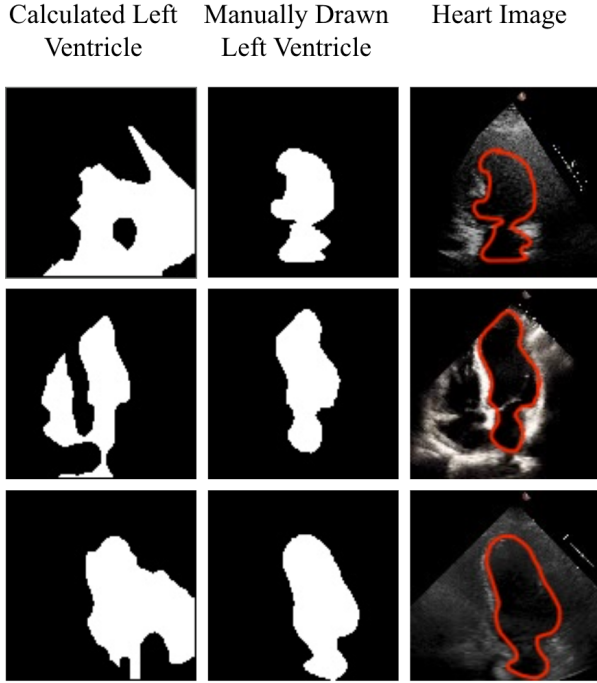


Fig. 12: Examples of Accidental Merging of the Calculated Left Ventricle with Background Components or the Right Ventricle

B. Mitral Valve Leaflets Identification

In order to evaluate the performance of the mitral valve detection algorithm, a more qualitative approach was necessary. Each video was broken down frame by frame. Each frame was then classified as one of the following three categories: Hit, Semi-Hit, Miss. In order to qualify as a hit, more than 75% of the mitral valve needed to have been labeled/highlighted. In order to qualify as a Semi-Hit, between 25% and 75% of the valve needed to have been highlighted, and in order to qualify as a Miss, less than 25% of the valve needed to have been highlighted. Fig. 13 demonstrates the shots that are seen as hits and misses to humans.

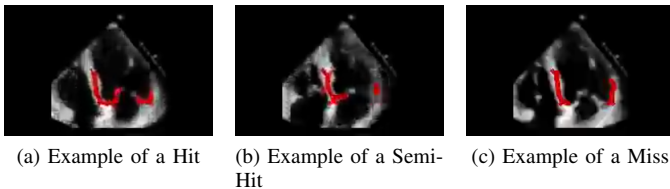


Fig. 13: Hand-drawn Ideal Contours and Maps

This required some degree of human judgment, which can of course lead to some human error. Therefore, this process was conducted on five different videos, and the results were tallied and displayed. The larger volume of data partially mitigates

the effect of human judgment error, making the data fairly reliable.

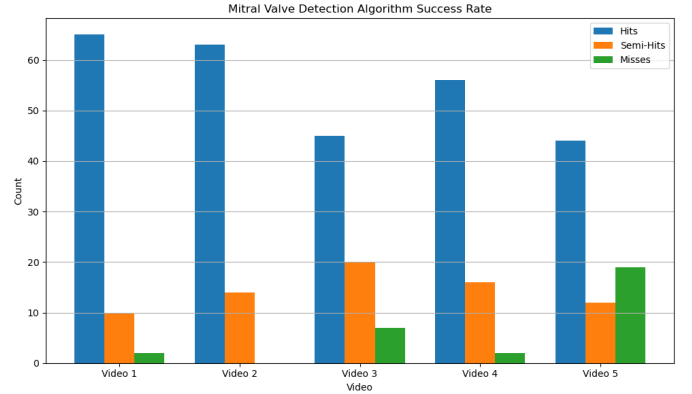


Fig. 14: Histogram of Hits, Semi-Hits, and Misses Across Five Experimented Videos

The result shows that our algorithm has a 92% accuracy on the identification of either leaflets of the mitral valve structures, and a 72.8% accuracy on the mitral valve structures if the user wants to identify both leaflets fully.

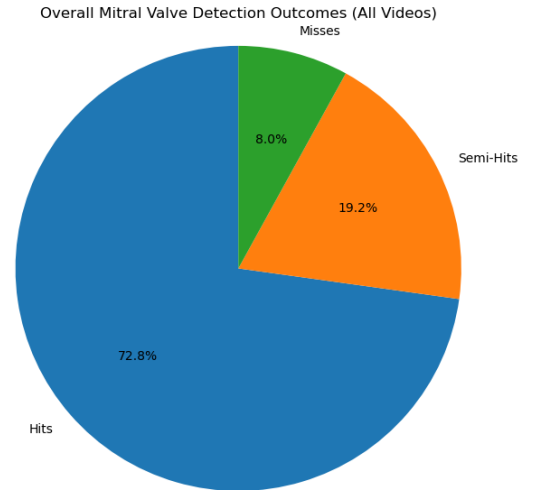


Fig. 15: Examples of Accidental Merging of the Calculated Left Ventricle with Background Components or the Right Ventricle

V. DISCUSSION

This project proposes a novel way that identifies dynamic features in an echocardiogram. Its code and data is included on this Github repository: <https://github.com/CooperX522/DSP-Final-Project-2025-Echocardiograms>. From the results demonstrated, one can conclude that this project's algorithms work well on identifying left ventricles with an 85% accuracy and mitral valve leaflets with a 92% accuracy. However, the scope of the algorithm is limited to only the Apical 4-Chamber view, and implements no prediction algorithm for cases where the

moving figure phases out of the frame. The Future work can be made to explore beyond these limits. Aside from this general picture, some detailed technical limits of either algorithm is discussed below.

A. LV Identification Limits and Future Direction

The issue of the left ventricle’s missing tip cannot be simply resolved by increasing the threshold value, as this would further weaken already faded walls. However, the missing tip could be captured by imposing stricter shape constraints on the ventricle. Once the left ventricle is identified, its shape could be remolded to better align with a standard ventricle template. Another approach is to apply a more sophisticated edge detection algorithm on the denoised image and create a binary mask from the detected edges. A common choice for this is the Sobel algorithm [10].

Resolving the missing lower portion of the left ventricle should be more straightforward. In most cases, both the upper and lower portions are present in the binary mask, and the challenge is accurately merging them. This could be achieved by estimating the orientation of the upper portion and using it to predict the location of the lower portion. Additionally, the valve itself could be detected, particularly in the Stanford heart videos where it is visible. Using the centroid of the upper portion of the ventricle along with the valve location, an accurate prediction of the lower portion’s position can be made.

The most challenging problem is the incorrect merging of the left ventricle with neighboring components due to weak or missing walls. This could be addressed by predicting the location of walls based on “clipped edges”—regions where walls fade and leave pointy bright objects. Two clipped edges facing each other likely indicate a faded wall, so a barrier can be constructed between these points to separate components. Despite these limitations, the algorithm performed relatively well. Several subtle design choices made a significant difference. For example, the vertical Gaussian blur, while not solving all issues, clarified the walls and improved component contours. Additionally, square-rooting the values in the intensity map for the watershed algorithm was essential, as it reduced the number of layers and decreased susceptibility to noise that often caused components to leak into neighboring regions. Outlier components were removed because their extreme error values skewed normalization, allowing for more accurate weight assignments when identifying the left ventricle. Finally, calculating the shape error using an iterative contour-fitting approach proved more effective than a correlation-based approach. Many background objects approximated the overall oval shape of the ideal ventricle, whereas real left ventricles often exhibited bending or slanting. The contour-fitting method emphasized the ventricle’s paddle shape, particularly the central notch, allowing even distorted ventricles to achieve lower shape error than background components.

Neural networks also present a promising alternative for left ventricle identification, especially given the abundance of available heart image data. A neural network could be

trained on a more diverse set of images, potentially making it more versatile and robust than the purely algorithmic approach implemented in this project.

B. MV Leaflets Identification Limits and Future Direction

The mitral valve leaflet identification results demonstrate that motion-based signal processing provides a robust cue for detecting valve dynamics in echocardiogram videos. The high accuracy achieved in identifying at least one leaflet (92%) suggests that temporal motion energy is a reliable indicator of mitral valve activity across cardiac cycles. This is consistent with the physiological behavior of the mitral valve, whose rapid opening and closing generates strong, spatially localized motion patterns relative to surrounding cardiac structures.

However, fully identifying both mitral valve leaflets remains more challenging, as reflected by the lower accuracy (72.8%) when strict criteria are applied. One major limitation arises from asymmetry in leaflet visibility. In many frames, one leaflet exhibits stronger motion contrast due to imaging angle, acoustic shadowing, or partial occlusion by surrounding anatomy. Because the proposed algorithm relies on motion magnitude and spatial coherence rather than explicit anatomical modeling, it may preferentially capture the more prominent leaflet while underrepresenting the second.

Another contributing factor is the variability in leaflet separation throughout the cardiac cycle. During phases when the valve is nearly closed, the two leaflets move in close proximity, often appearing as a single merged structure in the motion-supported mask. In such cases, the algorithm correctly identifies valve activity but lacks sufficient spatial resolution to distinguish individual leaflet boundaries. Conversely, during rapid opening phases, motion blur and speckle noise can cause fragmented detections, leading to partial labeling and classification as semi-hits.

The reliance on qualitative human judgment for evaluation also introduces uncertainty. Although categorizing frames as Hit, Semi-Hit, or Miss reflects clinical intuition, these categories are inherently subjective and may vary between observers. While the use of multiple videos partially mitigates this limitation, future work would benefit from more objective metrics, such as temporal consistency of leaflet trajectories or overlap with expert-annotated valve contours.

Several directions exist for improving mitral valve leaflet identification. First, incorporating temporal continuity constraints—such as tracking leaflet motion across frames using optical flow or temporal smoothing—could reduce frame-to-frame fragmentation and improve full leaflet reconstruction. Second, introducing geometric priors on expected leaflet orientation and separation could help disambiguate cases where both leaflets appear as a single motion cluster. Finally, adaptive region-of-interest refinement based on the previously detected left ventricle and valve annulus could further suppress false positives and improve robustness to imaging variability.

Despite these limitations, the proposed approach demonstrates that classical signal processing techniques, without reliance on annotated training data or learning-based models,

can achieve strong performance in identifying mitral valve dynamics. This highlights the potential for lightweight, interpretable DSP pipelines to assist clinicians in echocardiographic analysis, particularly in settings where data annotation or computational resources are limited.

REFERENCES

- [1] El Sabbagh A, Reddy YNV, Nishimura RA. Mitral Valve Regurgitation in the Contemporary Era: Insights Into Diagnosis, Management, and Future Directions. *JACC Cardiovasc Imaging*. 2018 Apr; 11(4):628-643. doi: 10.1016/j.jcmg.2018.01.009. PMID: 29622181.
- [2] Katz WE, Conrad Smith AJ, Crock FW, Cavalcante JL. Echocardiographic evaluation and guidance for MitraClip procedure. *Cardiovasc Diagn Ther*. 2017 Dec;7(6):616-632. doi: 10.21037/cdt.2017.07.04. PMID: 29302467; PMCID: PMC5752829.
- [3] Stanford AIMI, EchoNet-Dynamic, <https://stanfordaimi.azurewebsites.net/>
- [4] Ouyang D et al., EchoNet-Dynamic Scripts, <https://github.com/echonet/dynamic/>
- [5] GeeksforGeeks, "Spatial filters – averaging filter and median filter in image processing," Jul. 12, 2025. [Online]. Available: <https://www.geeksforgeeks.org/python/spatial-filters-averaging-filter-and-median-filter-in-image-processing/>
- [6] National Institutes of Health, "Filters – spatial Gaussian blur," MIPAV PubWiki. Available: https://mipav.cit.nih.gov/pubwiki/Filters_Spatial_Gaussian_Blur.html
- [7] GeeksforGeeks, "Image segmentation with watershed algorithm – OpenCV Python," Jul. 23, 2025. [Online]. Available: <https://www.geeksforgeeks.org/image-segmentation-with-watershed-algorithm-opencv-python/>
- [8] GeeksforGeeks, "Python OpenCV – distanceTransform() function," Jul. 23, 2025. [Online]. Available: <https://www.geeksforgeeks.org/python/python-opencv-distancetransform-function/>
- [9] NumPy Developers, "numpy.gradient," NumPy v2.1 documentation. [Online]. Available: <https://numpy.org/doc/2.1/reference/generated/numpy.gradient.html>
- [10] GeeksforGeeks, "Sobel edge detection vs. Canny edge detection in computer vision," Aug. 1, 2025. [Online]. Available: <https://www.geeksforgeeks.org/computer-vision/sobel-edge-detection-vs-canny-edge-detection-in-computer-vision/>